

Rawat Padungkiettiwong

DATA SCIENTIST

Bangkok, Thailand

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Education

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Aug 2020 - Aug 2024

- GPA: 3.61/4.3 (Converted Percentage: 93.10/100)
- Minor in Industrial and Systems Engineering
- Semiminor in Artificial Intelligence
- Relevant Coursework:

Computing: Deep Learning, Machine Learning with Graphs, Natural Language Processing, Computer Vision, Statistical Machine Learning, Data-Driven Decision Making and Control, Artificial Intelligence, Data Mining, Blockchain Security, Database Systems, Data Structures and Algorithms, Discrete Mathematics

Mathematics: Linear Algebra, Optimization, Stochastic Models, Bayesian Statistics, Probability and Statistics, Real and Complex Analysis, Differential Equations, Calculus

Kamnoetvidya Science Academy (KVIS)

Rayong, Thailand

HIGH SCHOOL DIPLOMA

May 2017 - Mar 2020

- Mathematics Concentration
- Graduation Research Project: "Ruin Probability with a Constant Premium Rate in a Financial Corporation Model"

Experience

Siam Commercial Bank

Bangkok, Thailand

PROFESSIONAL DATA SCIENTIST

Jul 2024 - Present

- Credit Risk
 - Developed behavioral scoring (B-Score) models for non-retail small and medium enterprise (SME) portfolios by engineering highly explainable feature pipelines, evaluating and benchmarking LightGBM against TabPFN architectures for tabular data.
 - Implemented a structured, vectorless Retrieval-Augmented Generation (RAG) framework to classify and unify multi-source qualitative credit concern lists for non-retail SME portfolios, optimizing early-warning risk identification pipelines.
 - Designed an application scoring (A-Score) framework for unsecured retail SME cohorts, implementing a semi-supervised Class-Aware Doubly Robust (CADR) estimator to mathematically mitigate selection bias caused by non-random missing labels.
 - Conducted statistical monitoring and quantitative risk analysis on heterogeneous retail loan portfolios (mortgages and SMEs), tracking feature drift and model performance consistency across distinct asset segments.
- Balance Sheet Optimization
 - Formulated a constrained mathematical optimization framework to determine optimal interest rate strategies, maximizing deposit booking volume while minimizing cost structures under multi-variable financial constraints.
 - Engineered a sequential flow-of-funds tracking system for the retail segment to model capital migration, enabling early customer churn detection and algorithmic cross-product recommendation mapping.
 - Developed a retail customer churn predictive model using LightGBM, integrating multi-modal data streams including high-dimensional behavioral feature pipelines and time-varying macroeconomic indicators.
 - Architected an enterprise-level liquidity risk framework to predict large-scale capital outflows for corporate portfolios, combining trend-seasonality decomposition, Cox Proportional Hazards models, and ARIMA variants to quantify temporal risk over multi-horizon forecasts.
- Engineering Frameworks & Infrastructure
 - Conducted empirical research on the generalization of automated, model-agnostic feature selection filters and wrappers within highly volatile datasets, evaluating performance trade-offs of diverse search heuristics; architected an experimental framework treating wrapper selection as a combinatorial optimization problem utilizing Genetic Algorithms and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) with Margin optimized via information-theoretic fitness objectives (mRMR).
 - Translated research findings into production-grade Python libraries, engineering unified, automated feature pipelines supporting both Pandas and distributed PySpark DataFrame architectures.

Users & Information Lab, KAIST

Daejeon, South Korea

UNDERGRADUATE RESEARCHER

Dec 2023 - Jun 2024

- Developed an adversarial training framework to mitigate selection bias in multiple-choice large language model (LLM) evaluation, successfully surpassing the performance bounds of contemporary prior-estimation benchmarks.
- Investigated architectures for integrating LLMs with Graph Neural Networks, optimizing node embeddings and structural representations within text-attributed citation networks.

Hong Kong Applied Science and Technology Research Institute (ASTRI)

Hong Kong

FINTech RESEARCH INTERN

Jun 2023 - Aug 2023

- Architected and implemented a privacy-centric federated learning platform for the financial sector using Vue.js and RESTful APIs; integrated rigorous quality assurance protocols within a CI/CD pipeline to ensure robust system functionality.

Siam Commercial Bank

Bangkok, Thailand

DATA SCIENCE INTERN

Jan 2023 - Feb 2023

- Optimized campaign timing to maximize Click-Through Rate (CTR) uplift by developing an offline multi-armed bandit (MAB) model, experimenting with various algorithms—including context-free (Epsilon-Greedy, UCB, Thompson Sampling) and contextual (LinUCB) methods—to determine the optimal policy.

International Scholar and Student Services (ISSS), KAIST

Daejeon, South Korea

ACADEMIC TUTOR (COMPUTER SCIENCE)

Sep 2022 - Dec 2022

- Provided individualized, one-on-one tutoring sessions in discrete mathematics for undergraduate students, facilitating targeted content revision and mentoring students on advanced problem-solving techniques.

Siam Commercial Bank

Bangkok, Thailand

DATA SCIENCE INTERN

Jun 2022 - Aug 2022

- Engineered a reinforcement learning-based decision engine for the Personalized Ad Recommendation (PAR) system using the Q-learning algorithm; mapped customer state transitions to optimize ad selection and maximize banking profitability through targeted delivery.

Projects

Mitigating Selection Bias in Multiple-Choice Question Answering with Novel Calibration for Large Language Models

INDEPENDENT RESEARCH

Feb 2024 - Jun 2024

- Formulated a novel post-hoc calibration framework to diminish position and token selection bias inherent in multiple-choice question answering tasks when employing LLMs.
- Designed optimization boundaries to surpass contemporary prior-estimation benchmarks, enhancing model robustness, fairness, and reasoning consistency without parameter fine-tuning.

Efficient Text-Attributed Graph Node Classification with Correct and Smooth

COURSEWORK PROJECT

Apr 2024 - May 2024

- Developed *ReCaS*, a novel graph representation learning approach that integrates the state-of-the-art TAPE framework with modified Correct and Smooth (C&S) label propagation heuristics.
- Augmented node feature representations and optimized error-correction steps, proving that simpler models, such as Multi-layer Perceptrons, can match or exceed baseline performance with reduced training time and fewer parameters.

Multiple-Choice Question Answering with Maieutic Prompting

COURSEWORK PROJECT

Sep 2023 - Dec 2023

- Extended the Maieutic Prompting paradigm to formalize logical consistency and truth-tree generation inside LLM reasoning pipelines for multiple-choice question answering.
- Introduced pairwise abductive reasoning mechanisms and logical integrity constraints to systematically resolve multi-choice label ambiguities, demonstrating competitive performance against standard Chain-of-Thought (CoT) baselines.

Timing Optimization Using Multi-Armed Bandit

INTERNSHIP RESEARCH PROJECT

Jan 2023 - Feb 2023

- Engineered an offline mathematical simulation framework to evaluate timing optimization strategies using stochastic and contextual multi-armed bandit formulations.
- Implemented and benchmarked context-free heuristics (ϵ -Greedy, Upper Confidence Bound, Thompson Sampling) alongside contextual formulations (Linear Upper Confidence Bound algorithm) to maximize expected click-through rate (CTR) uplift.

Improving Deep Semantic Dictionary Learning for Multi-label Image Classification

COURSEWORK PROJECT

Sep 2022 - Dec 2022

- Enhanced the classification accuracy of the Deep Semantic Dictionary Learning framework by optimizing the underlying visual feature learning pipelines and the Autoencoder-driven semantic dictionary generation layers.

State-based Propensity: Personalized Ad Recommendation System

INTERNSHIP RESEARCH PROJECT

Jun 2022 - Aug 2022

- Constructed a sequential decision-making propensity framework utilizing deep reinforcement learning paradigms to optimize targeted item selections.
- Implemented a Q-learning agent to model customer state transitions based on historical interaction vectors, maximizing long-term expected reward signals and conversion propensity.

Honors & Awards

2023	Encouragement Award (Ranked 3rd in Model Performance) , The 2nd KAIST-POSTECH-UNIST Data Science Contest	South Korea
2020	Best Social Inspiration Award , The 16th International Students' Science Fair (ISSF)	Thailand
2017	Bronze Medal , China Southeastern Mathematical Olympiad	China
2016	Bronze Medal , International Mathematics Competition	Thailand
2016	Bronze Medal , Thailand Mathematical Olympiad	Thailand

Extracurricular Activities

Google Cloud AI Study Jam: #ChaiyoGCP Season 6 Bangkok, Thailand
PARTICIPANT — ACHIEVED TIER 2 (HIGHEST COMPLETION MILESTONE RANK) Mar 2026 - Apr 2026

- Completed a rigorous series of cloud computing labs focused on distributed AI infrastructure, deploying generative models using Vertex AI and developing LLM orchestrations via the Agent Development Kit.
- Earned multiple verified technical badges across cloud architecture, security, and machine learning pipelines, finishing within the highest designated participant milestone tier.

KAIST Algorithmic Problem Solving Club Daejeon, South Korea
MEMBER Sep 2021 - Dec 2021

- Engaged in weekly competitive programming sessions, analyzing and solving complex algorithmic challenges utilizing advanced data structures to optimize execution runtime.
- Conducted independent study into advanced algorithmic paradigms beyond the standard undergraduate curriculum, focusing on asymptotic analysis and computational complexity reduction.

Skills

Programming Languages	Python, Scala, R, SQL, Java, C++, C, Rust, Haskell, JavaScript
Frameworks & Libraries	PySpark, Scikit-learn, LightGBM, MLflow, PyTorch, Pandas, NumPy
Tools & Platforms	Databricks, Git, Docker, Linux

Prerequisite Course Mapping

As required by the admissions guidelines, the table below maps the foundational computer science prerequisites specified by UT Austin to my equivalent coursework and grades completed at KAIST.

UT Austin Prerequisite Course	KAIST Equivalent Course(s)	Grade
Discrete Math for Computer Science (CS 311)	Discrete Mathematics (CS20004/CS204)	A-
Introduction to Programming (CS 312)	Introduction to Programming (CS10001/CS101)	B+
Data Structures (CS 314)	Data Structures (CS20006/CS206)	B+
Algorithms and Complexity (CS 331)	Introduction to Algorithms (CS30000/CS300)	B+
Computer Organization and Architecture (CS 429)	Computer Organization (CS30101/CS311)	A-
Principles of Computer Systems (CS 439)	System Programming (CS20300/CS230) Operating Systems and Lab (CS30300/CS330)	B- B0